PathAI Engine - Assessment Report

Open University Learning Analytics Dataset (OULAD) | Engagement scoring, Week 6 risk prediction, and course recommendation

This report summarises a production-oriented PathAI prototype built to help universities understand engagement trajectories, identify students needing support by Week 6, and recommend suitable next modules. Each task produces an actionable product output rather than a standalone research result.

Task 1 - Behavioral Engagement Scoring

6	7	98.4% -> 19.0%
feature buckets	student archetypes	risk gradient from lowest to highest score band

The weekly 0-100 score uses activity volume, consistency, recency gaps, resource diversity, course-pace alignment, and assessment behavior. Archetypes translate those signals into recognizable patterns such as steady engagers, early dropouts, late recoverers, burst engagers, procrastinators, compliance engagers, and opportunistic engagers.

Task 2 - Disengagement Prediction And Staff Alerts

0.991	0.870	0.793	0.848
high-touch precision	ROC-AUC	top-60% F1	top-60% recall

The selected model is XGBoost, trained only on Week <= 6 features. The headline result is a strong ranking model (ROC-AUC 0.870) converted into three operational tiers: a precise high-touch advisor queue, broader light-touch behavioral nudges, and passive monitoring.

Task 3 - Course Recommendations

0.354	0.468	7/7	AAA, GGG, EEE
content hit@3	CF hit@3	catalog coverage	cold-start modules

The recommender compares collaborative filtering against a shared feature-space content model. The content model matches student Week 6 profiles to course profiles using cosine similarity, blended with a Wilson lower-bound pass-rate prior for a conservative cold-start solution.

Overall result: the prototype is credible enough for a 90-day decision-support pilot. The strongest product insight is how model outputs are converted into workflows that advisors can act on without creating alert fatigue.

What This Ships Into Product

Student engagement view. A weekly 0-100 trajectory with an archetype label and the strongest behavioral signals behind the score. This lets administrators separate low grades from true disengagement.

Advisor risk queue. A ranked Week 6 intervention list split into high-touch support, light-touch nudges, and monitoring. This turns model risk into an operational workflow instead of a raw probability column.

Next-module guidance. Top-3 course recommendations with conservative cold-start defaults for students with no history. This gives students and advisors a concrete planning output for the next semester.

Primary caveat: the prototype is evaluated on historical OULAD outcomes, so the next 90 days should prioritize live advisor feedback, subgroup calibration, and recommendation constraints such as timetable availability.

Data Cleaning And Archetypes

The enrollment grain is id_student x code_module x code_presentation. Student ID alone is not unique in OULAD. Cleaning decisions preserve behavior signal while preventing target leakage.

Check	Issue Count	Policy
studentInfo key uniqueness	0	Use student-module-presentation as the enrolment key.
studentRegistration key uniqueness	0	Use student-module-presentation as the enrolment key.
Unregistered enrolments not labelled Withdrawn	9	Audit only; official final_result remains the supervised target.
Withdrawn enrolments without unregistration date	93	Do not use date_unregistration as a predictive feature because it can occur after Week 6.
Missing score rows	173	Keep missing scores as missing; median-impute only inside model preprocessing.
Invalid score rows outside 0-100	0	Invalid values would be excluded from score aggregates.
Assessment rows with missing due date	11	Exclude missing-date assessments from weekly due/submission features.
Negative VLE activity dates (pre-start engagement)	688988	Valid pre-presentation activity; date is days before module start. Used as proactivity signal.
VLE activity after module presentation length	0	Exclude from weekly engagement panel.
Missing VLE week_from/week_to metadata	10486	Do not use week_from/week_to as model features.

Candidate Feature	Week 6 Available?	Used?	Reason
Pre-start VLE activity (negative dates)	Yes	Yes	Observed before day 0; reflects eagerness to begin course materials.
Week 1-6 VLE clicks	Yes	Yes	Observed before the Week 6 prediction point.
Week 1-6 active days	Yes	Yes	Observed before the Week 6 prediction point.
Week 1-6 activity diversity	Yes	Yes	Observed before the Week 6 prediction point.
Early non-exam assessment submissions	Yes	Yes	Only due/submitted work by Week 6 is used.
Student demographics/context	Yes	Yes	Known at registration or course enrolment.
Engagement score	Yes	Yes	Computed from Week 1-6 behavior using train-derived score weights.
Final result	No	Target only	Used only to train/evaluate the supervised label.

Negative VLE dates were retained as pre-start proactivity rather than dropped. This provides useful early signal for both risk and recommendation tasks.

Archetype	Behavioral Signature
Steady Engager	High mean engagement score, low score volatility, strong active-day consistency.
Early Dropout	Sharp negative early-to-late score delta, low late score, often withdrawn final result.
Late Recoverer	Large positive score delta from early weeks to late weeks.
Sporadic / Burst Engager	High score range, high max score, and several zero-click weeks.
Perfectionist Procrastinator	Successful/high-scoring enrolment with late submissions and lower punctuality.
Surface Level / Compliance Engager	High assessment completion but low activity diversity and low material-study frequency.
Opportunistic Engager	High material-click share and engagement spikes, but weaker active-week consistency.

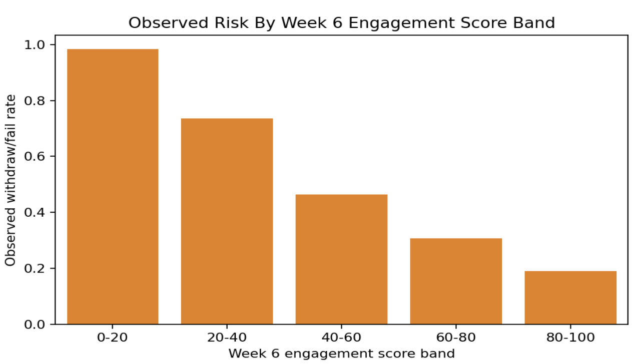
Task 1: Engagement Score

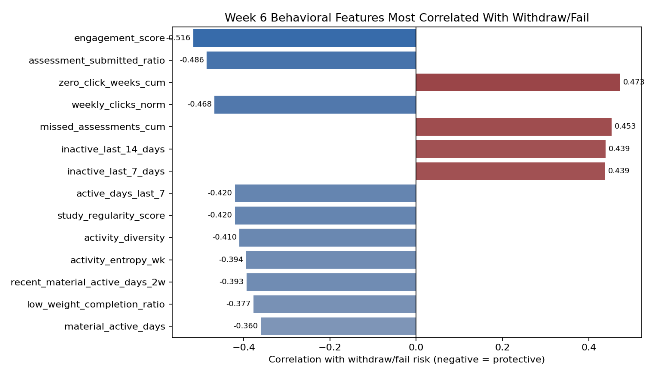
Approach: construct weekly behavioral features, normalize by module-presentation-week peers, and aggregate with train-derived weights. Components with weak train signal are assigned zero weight.

Component	Success AUC	Final Weight
Peer-normalized click volume	0.774	15%
Module-material study frequency	0.741	15%
Activity diversity	0.762	15%
Recent activity	0.773	15%
Pre-start course material engagement	0.690	10%
Active-day consistency	0.778	10%
Assessment completion	0.718	10%
Low-weight assignment completion	0.684	10%
Three-week trend	0.481	0%
Assessment punctuality	0.469	0%

Feature	Correlation With Withdraw/Fail
engagement_score	-0.516
assessment_submitted_ratio	-0.486
zero_click_weeks_cum	+0.473
weekly_clicks_norm	-0.468
missed_assessments_cum	+0.453
inactive_last_14_days	+0.439

Key decision: prioritize recency, consistency, diversity, and assessment follow-through over raw click count. This keeps the score transparent and resistant to click-spam behavior.





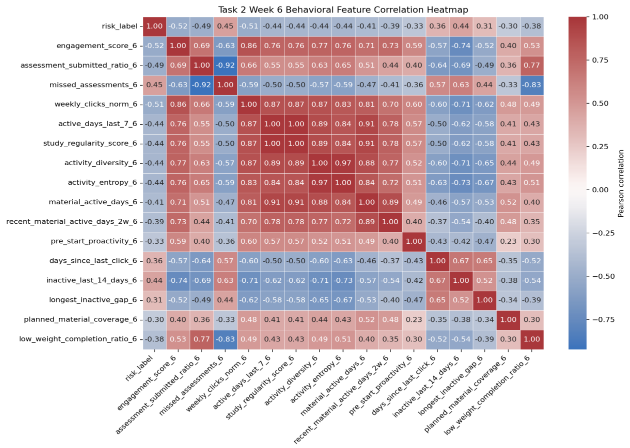
Task 2: Week 6 Disengagement Model

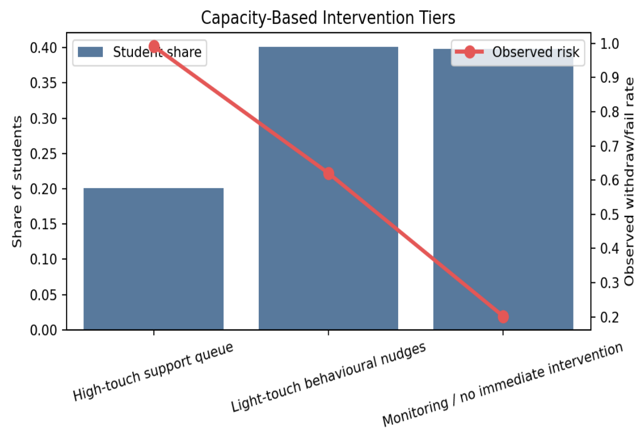
Approach: leakage-safe Week 6 binary classification. XGBoost was selected from a reproducible comparison set and then translated into advisor-capacity intervention tiers.

Model	Precision	Recall	F1	ROC-AUC
xgboost	0.738	0.842	0.787	0.863
xgboost_regularized	0.753	0.817	0.784	0.858
random_forest	0.754	0.816	0.783	0.857
logistic_regression	0.773	0.794	0.783	0.859

Cutoff	Precision	Recall	F1	Queue Share
High-touch 20%	0.991	0.377	0.546	20.1%
Top-60% light-touch	0.744	0.848	0.793	60.2%

Key decision: use tiered intervention (high-touch, light-touch, monitoring) instead of a single campus-wide alert to reduce advisor fatigue.





Task 3: Recommendation Engine

Approach: compare collaborative filtering to a shared feature-space content model. Student Week 6 vectors are matched to module profile vectors via cosine similarity, blended with a Wilson lower-bound pass prior.

Metric	Result
Evaluation split	Train: 2013B/2013J/2014B, Holdout: 2014J
Holdout students	1647
Content hit@3	0.354
Collaborative hit@3	0.468
Content coverage	7/7
Collaborative coverage	7/7
Cold-start modules	AAA, GGG, EEE

Actual Module	Content hit@3	CF hit@3
AAA	0.027	0.027
BBB	0.333	0.400
CCC	0.519	0.757
DDD	0.046	0.029
EEE	0.298	0.143
FFF	0.142	0.065
GGG	0.000	0.000

Key decision: replace raw pass-rate priors with Wilson lower-bound priors to reduce small-sample bias. This makes content recommendations more stable and defensible.

Gaps And 90-Day Plan

This prototype is intentionally simple and explainable, but not yet production-complete. The table below lists key gaps and concrete next actions.

Gap	Why It Matters	90-Day Action
Limited temporal depth	One-semester windows can miss drift	Move to rolling-semester retraining and drift checks
No advisor feedback loop	Model thresholds are not tied to intervention outcomes	Capture advisor action/outcome labels and recalibrate monthly
Recommendation label weakness	Next module may reflect timetable constraints	Add schedule constraints and student intent tags
Subgroup calibration not fully audited	Uneven risk quality can harm trust	Run subgroup calibration audits and apply targeted recalibration
Reproducibility hardening	Notebook and production states can diverge	Introduce versioned feature snapshots and run manifests

Phase	Delivery Focus
Days 0-30	Deploy advisor dashboard in shadow mode; validate data freshness and triage quality
Days 31-60	Tune intervention thresholds by module and staffing constraints
Days 61-90	Retrain with feedback labels; publish impact and calibration report

Why this is not best-in-class yet: it optimizes predictive performance on available labels, but does not yet optimize intervention utility, fairness constraints, and real scheduling feasibility end-to-end.